

LIMESTONE 4 NNW, ME, USA

WMO: 727100

Lat: 46.9601N

Lon: 67.8833W

Elev: 737

StdP: 14.31

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 94645

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.6	-10.7	-20.8	1.8	-13.1	-16.6	2.3	-8.6	17.7	23.7	14.6	19.9	2.0	N/A	0.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	82.5	67.3	79.5	65.7	76.7	64.5	70.3	77.9	68.6	75.7	67.2	74.1	6.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.9	105.3	73.3	66.1	99.0	72.2	64.5	93.3	70.8	34.6	77.9	33.2	75.9	32.0	73.8	
(p)															

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	10.8	8.9	-24.2	88.1	5.4	2.5	-28.1	89.9	-31.2	91.3	-34.3	92.7	-38.1	94.5		
(5)			-23.1	73.1	4.1	1.1	-26.0	73.9	-28.4	74.5	-30.7	75.1	-33.6	75.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.1	10.3	12.8	22.8	36.9	50.3	59.1	65.1	63.2	55.2	43.3	30.9	17.8		
	DBStd	21.04	12.92	11.49	10.84	8.05	7.21	6.46	5.16	5.43	7.16	7.83	9.07	11.20		
	HDD50	5440	1231	1041	843	399	84	4	0	0	28	239	573	998		
	HDD65	9585	1696	1461	1307	844	458	199	62	99	301	673	1022	1463		
	CDD50	1477	0	0	1	5	95	277	469	411	186	31	2	0		
	CDD65	145	0	0	0	0	4	22	66	44	9	0	0	0		
	CDH74	1226	0	0	0	2	81	228	519	309	83	4	0	0		
	CDH80	219	0	0	0	0	19	46	97	46	11	0	0	0		
	WSAvg	3.8	4.1	4.3	4.3	4.3	4.1	3.5	3.2	3.1	3.3	3.8	4.1	3.9		
	PrecAvg	45.6	3.6	2.7	3.4	3.1	3.8	4.3	4.4	4.0	4.0	4.4	3.7	4.2		
(15) Precipitation	PrecMax	58.3	6.1	5.0	6.7	5.2	6.1	8.3	7.3	7.0	8.2	8.3	7.8	7.4		
	PrecMin	30.9	1.4	0.9	1.3	1.3	1.3	2.4	1.5	1.7	1.6	1.3	1.6	1.3		
	PrecStd	5.9	1.3	0.9	1.3	1.0	1.2	1.5	1.6	1.5	1.7	1.9	1.5	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.1	43.1	53.3	68.4	82.5	85.2	86.4	84.5	81.2	72.3	59.7	47.4		
		MCWB	44.0	40.2	47.6	50.5	64.7	65.9	68.6	69.1	67.1	60.5	57.8	46.8		
	2%	DB	38.7	36.8	44.5	60.1	74.8	80.1	82.7	80.6	76.2	64.7	53.0	40.9		
		MCWB	37.0	33.0	38.2	49.2	58.3	63.3	67.7	67.1	66.0	58.0	49.1	39.6		
	5%	DB	34.3	33.5	40.6	54.7	69.3	76.0	79.6	77.4	72.0	60.2	48.3	36.8		
		MCWB	32.8	31.0	35.4	45.9	54.9	62.2	66.5	65.4	62.7	55.3	44.8	35.2		
	10%	DB	30.3	30.7	37.4	50.1	64.7	72.2	76.7	74.5	68.5	56.5	44.4	33.5		
		MCWB	28.6	28.6	33.2	42.6	53.3	59.3	65.2	63.9	61.9	51.9	41.4	32.0		
	0.4%	WB	44.6	42.5	51.7	56.5	67.0	69.4	73.5	71.7	71.0	62.6	56.7	47.8		
		MCDB	45.0	44.0	60.2	61.4	78.8	77.8	82.3	80.8	80.0	65.8	58.5	48.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.5	34.7	41.1	50.8	62.1	66.7	70.8	69.4	68.0	59.9	49.7	41.2		
		MCDB	38.8	36.6	43.0	55.4	69.6	73.8	78.1	76.4	73.7	63.3	51.4	42.0		
	5%	WB	32.8	31.7	36.0	46.6	58.7	64.2	69.1	67.8	65.8	56.7	44.6	36.7		
		MCDB	34.4	33.8	39.6	53.6	64.3	71.2	76.3	74.7	70.7	60.0	46.6	37.7		
	10%	WB	28.8	28.7	33.0	42.8	55.6	61.6	67.7	66.2	63.5	52.9	41.2	33.0		
		MCDB	30.5	30.7	36.5	48.0	62.8	68.3	75.0	72.3	69.1	56.4	44.1	34.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.9	19.0	18.6	17.9	20.4	19.7	19.5	19.7	20.4	16.6	13.5	15.2		
		MCDBR	17.4	17.9	20.1	27.2	29.1	25.5	23.1	23.3	24.4	23.5	18.5	15.2		
		MCWBR	18.0	17.9	15.0	16.9	14.9	13.6	11.4	11.3	13.9	15.9	16.2	13.9		
		MCDBR	18.4	20.0	17.3	22.6	22.1	21.1	19.8	19.6	20.3	18.5	17.0	15.0		
	5% WB	MCWBR	18.5	18.5	14.6	16.8	14.1	13.1	11.0	10.9	13.4	15.0	16.1	15.3		
(40) Clear-Sky Solar Irradiance	taub		0.253	0.270	0.302	0.341	0.354	0.374	0.405	0.385	0.356	0.319	0.291	0.272		
	taud		2.353	2.287	2.287	2.375	2.413	2.355	2.291	2.347	2.418	2.524	2.524	2.385		
	Ebn at Noon		270	287	292	290	289	282	271	272	271	268	255	247		
	Edh at Noon		22	30	35	36	36	38	40	36	31	24	19	20		
(44) All-Sky Solar Radiation	RadAvg		444	725	1102	1409	1590	1725	1722	1512	1191	714	405	311		
	RadStd		55	81	76	127	164	119	147	95	90	65	49	29		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page